

Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 4A nr. 52.4

$$\begin{aligned} & \int \sin^4 x \cos^2 x dx \\ &= \int \left(\frac{1 - \cos 2x}{2} \right)^2 \left(\frac{1 + \cos 2x}{2} \right) dx \\ &= \frac{1}{8} \int (1 - 2 \cos 2x + \cos^2 2x) (1 + \cos 2x) dx \\ &= \frac{1}{8} \int (1 - \cos 2x - \cos^2 2x + \cos^3 2x) dx \\ &= \frac{1}{8} \int \left(1 - \cos 2x - \frac{1 + \cos 4x}{2} + \cos 2x \frac{1 + \cos 4x}{2} \right) dx \\ &= \frac{1}{8} \int \left(1 - \cos 2x - \frac{1}{2} - \frac{1}{2} \cos 4x + \frac{1}{2} \cos 2x + \frac{1}{2} \cos 2x \cos 4x \right) dx \\ &= \frac{1}{8} \int \left(\frac{1}{2} - \frac{1}{2} \cos 2x - \frac{1}{2} \cos 4x + \frac{1}{2} \cos 2x \cos 4x \right) dx \\ &= \frac{1}{8} \int \left(\frac{1}{2} - \frac{1}{2} \cos 2x - \frac{1}{2} \cos 4x + \frac{1}{4} (\cos 6x + \cos 2x) \right) dx \\ &= \frac{1}{8} \int \left(\frac{1}{2} - \frac{1}{4} \cos 2x - \frac{1}{2} \cos 4x + \frac{1}{4} \cos 6x \right) dx \\ &= \int \left(\frac{1}{16} - \frac{1}{32} \cos 2x - \frac{1}{16} \cos 4x + \frac{1}{32} \cos 6x \right) dx \\ &= \frac{x}{16} - \frac{1}{32} \frac{\sin 2x}{2} - \frac{1}{16} \frac{\sin 4x}{4} + \frac{1}{32} \frac{\sin 6x}{6} + C \\ &= \frac{x}{16} - \frac{\sin 2x}{64} - \frac{\sin 4x}{64} + \frac{\sin 6x}{192} + C \end{aligned}$$

References